Lecture 03, Math 441

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Integrable Equations

Ex: $y' = x^3$ so $y' = f(x)$

Just integrate in x: $y = \frac{x^4}{4} + A$

where A (constant of integration) is a free parameter.

In general: $y' = f(x) \rightarrow y(x) = \int f(x)\mathrm{d}x$

Another notation:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x^3 \rightarrow \mathrm{d}y = x^3\mathrm{d}x \rightarrow \int \mathrm{d}y = \int x^3\mathrm{d}x \rightarrow y(x) = \frac{x^4}{4} + A$$

Higher order ex: $y'' = \cos(x)$. Just integrate twice:

$y' = \sin(x) + A \quad \rightarrow$ integrate again: $\quad y = -\cos(x) + Ax + B$

where A and B are free parameters.

Note: the general solution of a linear n-th order ODE has n free parameters.

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$$\frac{\mathrm{d}^2 y}{\mathrm{d}x}^2 = \cos x$$

DO NOT DO THIS: $\rightarrow \mathrm{d}^2 y = \cos(x)\mathrm{d}x^2$

INSTEAD DO THIS:

$\rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\frac{\mathrm{d}y}{\mathrm{d}x}) = \cos(x) \quad \rightarrow \quad \int \mathrm{d}(\frac{\mathrm{d}y}{\mathrm{d}x}) = \int \cos(x)\mathrm{d}x$

$\rightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \sin x + A$

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Separable Equations

Ex: $\frac{dy}{dx} = 2xy$ is not integrable, but it is separable. Separate and then integrate:

$$\frac{dy}{dx} = 2xy \rightarrow \frac{dy}{y} = 2x dx \rightarrow \int \frac{dy}{y} = \int 2x dx \rightarrow \ln(y) = \frac{x^2}{2} + A$$

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In general, if $\frac{\mathrm{d}y}{\mathrm{d}x} = f(x,y)$ and $f(x,y) = g(x)h(y)$ then we can

separate and integrate

$$\frac{\mathrm{d}y}{\mathrm{d}x} = g(x)h(y) \rightarrow \frac{\mathrm{d}y}{h(y)} = g(x)\mathrm{d}x \rightarrow \int \frac{\mathrm{d}y}{h(y)} = \int g(x)\mathrm{d}x \cdots$$

Ex:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^2 + y}{y + 1} - 1 \rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^2 + y - y - 1}{y + 1} = \frac{x^2 - 1}{y + 1} = (x^2 - 1)\frac{1}{y + 1}$$

which is separable.

So separate and integrate:

$$(y + 1)\mathrm{d}y = (x^2 - 1)\mathrm{d}x \rightarrow \int (y + 1)\mathrm{d} = \int (x^2 - 1)\mathrm{d}x$$

$$\rightarrow \frac{y^2}{2} + y = \frac{x^3}{3} - x + A \text{ (implicit solution)}$$

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To find the explicit solution we rewrite:

$$y^2 + 2y - \frac{2}{3}x^3 + 2x + A = 0$$

(we keep using A, since it is a free undetermined parameter)

$$y(x) = -1 \pm \sqrt{1 + \frac{2}{3}x^3 - 2x - A} \qquad \text{(explicit solution)}$$

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To the determine the sign in front of the root we would need an IC.

For ex, if the IC is $y(0) = 3$ then we have

$$y(0) = -1 \pm \sqrt{1 - A} = 3 \implies \sqrt{1 - A} = 4$$

$$\implies 1 - A = 16 \implies A = -15$$

which means that $A = -15$ and we need to chose the + sign.

So the solution to the IVP would be

$$y(x) = -1 + \sqrt{1 + \frac{2x^3}{3} - 2x + 15}$$

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Ex: $\frac{\mathrm{d}y}{\mathrm{d}x} = y - 1 + x - xy = y - 1 - x(y-1) \rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = (y-1)(1-x)$

$\rightarrow \frac{\mathrm{d}y}{y-1} = (1-x)\mathrm{d}x$ assuming $y \neq 1$.

Integrate: $\int \frac{\mathrm{d}y}{y-1} = \int (1-x)\mathrm{d}x$

$$\rightarrow \ln(y-1) = x - \frac{x^2}{2} + A \rightarrow y - 1 = \mathrm{e}^{x - \frac{x^2}{2} + A} = C\mathrm{e}^{x - \frac{x^2}{2}}$$

$$\Longrightarrow \boxed{\mathrm{y(x)} = 1 + \mathrm{Ce}^{x - \frac{x^2}{2}}} \text{ where } C = \mathrm{e}^{A}.$$

Ex:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3y^2}{x^4} \;\rightarrow\; \int \frac{\mathrm{d}y}{y^2} = \int \frac{3}{x^4}\mathrm{d}x \;\text{ (assuming } y \neq 0)$$

$$\rightarrow\; -\frac{1}{y} = -\frac{1}{x^3} + A \;\rightarrow\; \frac{1}{y} = \frac{1}{x^3} + A = \frac{1 + Ax^3}{x^3}$$

(ignoring the sign of A since it is a free parameter)

And finally

$$y(x) = \frac{x^3}{1 + Ax^3} \qquad (y \neq 0)$$

AND $\boxed{\text{y(x)=0}}$ (singular solution)

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Singular Solutions

We assumed $y \neq 0$. But what if $y = 0$?

Let's try the function $y(x) = 0$. Notice that $y'(x) = 0$ and it verifies the original equation:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3y^2}{x^4} \rightarrow 0 = 0$$

So $y(x) = 0$ is a solution but it is not included in the general solution we found above, it is a singular solution.

Nonlinear equations may have singular solutions.

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Exact Equations

Rewrite the following equations using differentials:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^2-1}{y+1} \rightarrow (y+1)\mathrm{d}y = (x^2-1)\mathrm{d}x \rightarrow (1-x^2)\mathrm{d}x+(y+1)\mathrm{d}y = 0 \ (*)$$

Consider a function $\Psi(x,y)$ and take the differential:

$$\mathrm{d}\Psi = \frac{\partial\Psi}{\partial x}\mathrm{d}x + \frac{\partial\Psi}{\partial y}\mathrm{d}y$$

$$(1-x^2)\mathrm{d}x + (y+1)\mathrm{d}y = 0 \rightarrow \mathrm{d}\Psi = 0 \rightarrow \Psi(x,y) = A$$

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Compare $(1-x^2)\mathrm{d}x + (y+1)\mathrm{d}y$ with $\frac{\partial \Psi}{\partial x}\mathrm{d}x + \frac{\partial \Psi}{\partial y}\mathrm{d}y$

If

$$\frac{\partial \Psi}{\partial x} = (1-x^2)$$

and

$$\frac{\partial \Psi}{\partial y} = (y+1)$$

We can easily check that

$$\Psi(x, y) = x - \frac{x^3}{3} + \frac{y^2}{2} + y$$

has the correct partial derivatives.

Therefore the solution to the ODE can be written as

$$x - \frac{x^3}{3} + \frac{y^2}{2} + y = A$$

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When does this method work?

Rewrite a 1st order ODE as $M(x,y)\,\mathrm{d}x + N(x,y)\,\mathrm{d}y = 0$. and the

equation is equivalent to $\mathrm{d}\Psi = \frac{\partial \Psi}{\partial x}\mathrm{d}x + \frac{\partial \Psi}{\partial y}\mathrm{d}y = 0$

If we can find a function $\Psi(x,y)$ such that

$\frac{\partial \Psi}{\partial x} = M(x,y)$ and $\frac{\partial \Psi}{\partial y} = N(x,y)$

$\Longrightarrow$ $\Psi(x,y) = A$ is a solution

(and the equation is called exact)

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Condition for Exactness

Notice that if $M(x,y) = \frac{\partial \Psi}{\partial x}$ then $\frac{\partial M(x,y)}{\partial y} = \frac{\partial^2 \Psi}{\partial y \partial x} = \Psi_{xy}$

and if $N(x,y) = \frac{\partial \Psi}{\partial y}$ then $\frac{\partial N(x,y)}{\partial x} = \frac{\partial^2 \Psi}{\partial x \partial y} = \Psi_{yx}$

If these are continuous, then $\frac{\partial^2 \Psi}{\partial y \partial x} = \frac{\partial^2 \Psi}{\partial x \partial y}$

which means that

$$\frac{\partial M(x,y)}{\partial y} = \frac{\partial N(x,y)}{\partial x}$$

This is the condition for the equation to be exact.

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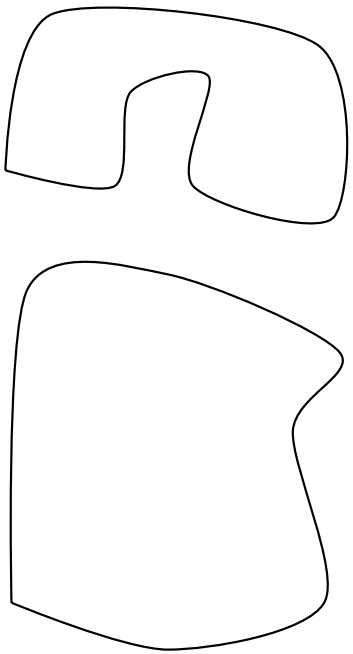
Theorem: The equation

$$M(x,y)\mathrm{d}x + N(x,y)\mathrm{d}y = 0$$

with M, N, M_y, N_x continuous in a simply connected region R, is an **exact equation** in R if and only if

$$\frac{\partial M(x,y)}{\partial y} = \frac{\partial N(x,y)}{\partial x}$$

examples of simply connected region R (has no holes)



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We already proved the "only if" part. Let's prove the "if" part.

I.e. Prove that if $\frac{\partial M(x,y)}{\partial y} = \frac{\partial N(x,y)}{\partial x} \rightarrow$ then $\exists$ a function $\Psi(x,y)$

such that $\frac{\partial \Psi}{\partial x} = M(x,y)$ and $\frac{\partial \Psi}{\partial y} = N(x,y)$

We prove this by finding Ψ.

Define $\Psi(x,y) = \int M(x,y)\mathrm{d}x + h(y)$

$\rightarrow \quad \Psi(x,y) = Q(x,y) + h(y)$

where we called $\int M(x,y)\mathrm{d}x = Q(x,y)$

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Now we need $\frac{\partial \Psi}{\partial y} = \frac{\partial Q(x,y)}{\partial y} + h'(y) = N(x,y)$

or $h'(y) = N(x,y) - \frac{\partial Q(x,y)}{\partial y}$ (*)

The right hand side cannot depend on x, so we need

$N_x - Q_{yx} = 0 \implies N_x - M_y = 0$

$\frac{\partial^2 Q}{\partial x \partial y} - \frac{\partial N(x,y)}{\partial x} = 0$ which is $\frac{\partial M(x,y)}{\partial y} - \frac{\partial N(x,y)}{\partial x} = 0$

This is true by the hypothesis.

In practice, we can use (*) to find $h(y)$ and therefore $\Psi(x,y)$.

Ex: $(x\cos(y)+\mathrm{e}^{y})\,y' = -2x-\sin(y)$

$\rightarrow (x\cos(y)+\mathrm{e}^{y})\,\mathrm{d}y = (-2x-\sin(y))\mathrm{d}x$

Rewrite as $(2x+\sin(y))\,\mathrm{d}x + (x\cos(y)+\mathrm{e}^{y})\,\mathrm{d}y = 0$

and call $M(x,y) = (2x+\sin(y))$ and $N(x,y) = (x\cos(y)+\mathrm{e}^{y})$

Check that $M_y = N_x$

$M_y = \cos(y) = N_x = \cos(y)$, it works so the equation is exact.

Let's find $\Psi(x, y)$. Start by setting

$$\Psi(x, y) = \int M \mathrm{d}x + h(y) = x^2 + x \sin(y) + h(y)$$

We need $\Psi_y = N(x, y)$

$$\Psi_y = x \cos(y) + h'(y) = x \cos(y) + \mathrm{e}^y$$

So $h'(y) = \mathrm{e}^y$ and integrating: $h(y) = \mathrm{e}^y$

Therefore $\Psi(x, y) = x^2 + x \sin(y) + \mathrm{e}^y$

$$\mathrm{d}\Psi = 0$$

$$\rightarrow \Psi = A \quad \rightarrow \boxed{x^2 + x \sin(y) + \mathrm{e}^y = A}$$

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